MINO PEPTIDES

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VOLUME TWO · THE COMPENDIUM

The Architecture

of a New Longevity

A field guide to the molecules quietly redrawing the boundary between aesthetics, recovery, and the science of how we age.

TWELVE COMPOUNDS · ONE STANDARD · 2026 EDITION

A note before you turn the page.

The most interesting science in longevity right now isn't loud. It's quiet, careful, and increasingly hard to ignore.

For most of the last decade, the conversation about aging has been dominated by two extremes — cosmetic intervention on one side, and aspirational supplementation on the other. Somewhere in the middle, almost without notice, a third category took shape. It is built around peptides: short, precise chains of amino acids that the body already speaks fluently, deployed with pharmaceutical-grade rigor and increasingly studied in respected clinical journals.

This is the territory Mino occupies. We supply licensed medical practitioners with European-manufactured, pharmaceutical-grade peptide compounds — every batch verified by HPLC and mass spectrometry to a minimum 99% purity, every product produced under EU-GMP Annex 1 standards. We do not sell to consumers. We do not retail online. We exist to give physicians, longevity clinicians, and aesthetic specialists a single source of truth for the molecules now shaping their practice.

Volume Two is not a catalogue. It is a curated reference to twelve of the most-studied compounds in our portfolio, written for the patient who wants to understand what is actually being prescribed for them — and the practitioner who wants a single document that holds together as both science and design. Every claim in these pages is anchored to peer-reviewed literature. Every compound is presented honestly: what is FDA-approved, what is internationally approved, what remains investigational, and what the published research actually shows.

What follows is the most carefully compiled introduction to the field we know how to make.

— Mino Editorial

The Twelve.

A CURATED FIELD OF COMPOUNDS

i.

Metabolic Reset

The single most consequential category in modern medicine — and the one most often misrepresented in the public conversation. Three compounds, each with published Phase 3 evidence, each working through a different door into the same biology.

INVESTIGATIONAL · PHASE 3

Mino GLP3-RT

RETATRUTIDE · TRIPLE AGONIST

The first molecule to engage three metabolic receptors simultaneously — and the one producing the largest weight reductions ever recorded in a published Phase 3 trial.

WHAT IT IS

GLP3-RT is the brand name under which Mino supplies retatrutide, a synthetic peptide that activates three receptors at once: GLP-1, GIP, and glucagon. It is the first triple agonist to advance to Phase 3 clinical trials in obesity. Eli Lilly is the originating developer, and the compound remains investigational; it has not been approved by the FDA. Mino supplies it exclusively to licensed medical practitioners.

WHAT THE RESEARCH SHOWS

In December 2025, Lilly reported topline results from TRIUMPH-4, a Phase 3 trial in adults with obesity. Participants on the highest dose lost an average of 71.2 pounds, with a placebo-adjusted reduction of 26.6%. These figures place retatrutide at the leading edge of weight-loss pharmacology to date. The trial also reported a transient dysesthesia signal that the FDA will evaluate during review.

CONSIDERED FOR

Adults with obesity or overweight with metabolic complications, in clinical and research settings under physician supervision. Practitioners typically reserve GLP3-RT for patients who have plateaued on existing GLP-1 therapy or who are seeking the most aggressive metabolic intervention currently available.

ORIGIN	MECHANISM	STATUS
Eli Lilly	*GLP-1 / GIP / Glucagon triple agonism*	*Phase 3 · Investigational*

FDA APPROVED · MOUNJARO · ZEPBOUND

Mino Tirzepatide

DUAL GIP / GLP-1 RECEPTOR AGONIST

The first dual incretin to win FDA approval — and the molecule that, more than any other single drug, changed how the public talks about weight, hunger, and the biology of appetite.

WHAT IT IS

Tirzepatide is a synthetic peptide that activates both the GIP and GLP-1 receptors. Developed by Eli Lilly, it received FDA approval in May 2022 for type 2 diabetes under the brand name Mounjaro, and a second FDA approval in November 2023 for chronic weight management under the brand name Zepbound. Mino supplies pharmaceutical-grade tirzepatide as a regulated peptide product to licensed medical providers.

WHAT THE RESEARCH SHOWS

The SURMOUNT-1 trial, published in the New England Journal of Medicine in 2022, reported mean weight reductions of approximately 21% in adults with obesity at the highest dose over 72 weeks. Subsequent SURMOUNT trials confirmed sustained efficacy across diverse populations and produced cardiovascular and sleep apnea benefits that have shaped the current standard of care.

CONSIDERED FOR

Adults with type 2 diabetes or chronic weight management indications, prescribed by physicians within FDA-approved labeling. Increasingly used in longevity-focused practices as a metabolic foundation alongside lifestyle protocols.

APPROVED	MECHANISM	PIVOTAL TRIAL
FDA · 2022 · 2023	*Dual GIP / GLP-1 incretin agonism*	SURMOUNT-1 · NEJM 2022

INVESTIGATIONAL · PHASE 3 REPORTED

Mino Cagrilintide

LONG-ACTING AMYLIN ANALOG

A different door into appetite biology — one that works through amylin rather than GLP-1 — and the cornerstone of Novo Nordisk's next-generation obesity program.

WHAT IT IS

Cagrilintide is a long-acting analog of amylin, a hormone co-secreted with insulin from the pancreas after meals. It targets amylin and calcitonin receptors in the brainstem and hypothalamus, slowing gastric emptying and producing satiety through pathways distinct from GLP-1 biology. Developed by Novo Nordisk, it remains investigational and is not yet FDA-approved as a monotherapy. Mino supplies it as a research-grade peptide to licensed practitioners.

WHAT THE RESEARCH SHOWS

The REDEFINE 1 trial, published in the New England Journal of Medicine in June 2025, evaluated cagrilintide both alone and in combination with semaglutide. The combination produced approximately 22.7% mean weight loss at 68 weeks. Cagrilintide as monotherapy produced 11.8% mean weight loss in a sub-analysis presented at EASD 2025. Novo Nordisk filed a New Drug Application for the combination product CagriSema in December 2025 and announced a dedicated Phase 3 monotherapy program called RENEW.

CONSIDERED FOR

Investigational use in clinical settings exploring complementary mechanisms to GLP-1 therapy. Of particular interest to providers managing patients with GLP-1 intolerance or those seeking alternative satiety pathways.

ORIGIN	MECHANISM	STATUS
Novo Nordisk	*Amylin / Calcitonin Receptor Agonism*	*NDA Filed · Phase 3*

ii.

Recovery & Repair

The two most-discussed regenerative peptides in the field — extensively studied in preclinical models, increasingly familiar in physician-supervised settings, and presented here with the honesty their evidence base requires.

PRECLINICAL · LIMITED HUMAN DATA

Mino BPC

BPC-157 · BODY PROTECTION COMPOUND

Originally isolated from a protein in human gastric juice — and now the most-discussed regenerative peptide of the decade.

WHAT IT IS

BPC-157 is a synthetic peptide derived from a fragment of body protection compound, a protein found in human gastric juice. Unlike many peptides that degrade rapidly in the digestive tract, BPC-157 is stable in gastric acid, which has made it a long-running subject of interest in tissue repair research. The compound is not FDA-approved. Mino supplies it exclusively as a research-grade peptide to licensed practitioners.

WHAT THE RESEARCH SHOWS

Preclinical research on BPC-157 spans more than two decades, with hundreds of published animal studies reporting effects on tendon, ligament, muscle, and gastrointestinal tissue repair. Proposed mechanisms include angiogenesis through VEGFR2 signaling, modulation of nitric oxide pathways, and stimulation of growth factor expression. Human clinical data remain limited; published pilot studies in humans are small in scale, and most evidence is preclinical.

CONSIDERED FOR

Investigational use in physician-supervised research settings exploring soft-tissue repair, tendon and ligament recovery, and gut-barrier protection. Honest framing matters: the preclinical signal is strong, the human evidence is still emerging.

DISCOVERED	MECHANISM	STATUS
1990s · *Sikiric et al.*	*Angiogenesis* · VEGFR2 · NO *pathway*	*Investigational · Research only*

PRECLINICAL · INVESTIGATIONAL

Mino TB

TB-500 · THYMOSIN BETA 4 FRAGMENT

A synthetic fragment of a protein the body already manufactures — studied for its role in cell migration, angiogenesis, and the choreography of tissue repair.

WHAT IT IS

TB-500 is a synthetic peptide based on a region of the naturally occurring protein thymosin beta 4. Native thymosin beta 4 is one of the most abundant proteins in cells and is involved in actin sequestration, cell migration, and wound repair. TB-500 is not FDA-approved. Mino supplies TB-500 as a research-grade peptide to licensed practitioners.

WHAT THE RESEARCH SHOWS

Preclinical studies in animal models report effects on cardiac repair, dermal wound healing, and corneal tissue regeneration, with proposed mechanisms involving actin binding, endothelial cell migration, and angiogenesis. Native thymosin beta 4 has been studied in early-phase human trials for ocular and cardiac indications. Human clinical data on TB-500 specifically remain limited, and most published evidence comes from animal and in vitro models.

CONSIDERED FOR

Investigational use in research settings examining soft-tissue recovery and connective tissue repair. Often paired with BPC-157 in physician-supervised protocols, though combination data are limited to clinical observation.

PARENT PROTEIN	MECHANISM	STATUS
Thymosin Beta 4	*Actin sequestration · Cell migration*	*Investigational · Research only*

iii.

Immune Architecture

A single compound — but the most clinically validated peptide in the entire portfolio. Approved internationally, recognized by the FDA for orphan indications, and the closest thing the field has to a gold standard for immune-modulating biology.

APPROVED IN 35+ COUNTRIES · 4 FDA ORPHAN DESIGNATIONS

Mino Thymosin Alpha

THYMOSIN ALPHA 1 · TA1 · ZADAXIN

The most clinically established immune-modulating peptide in the field — studied across more than thirty trials and approved internationally as Zadaxin for hepatitis B and C.

WHAT IT IS

Thymosin alpha 1 is a synthetic peptide identical to a naturally occurring fragment isolated from thymosin fraction 5, originally extracted from the thymus gland. It is marketed internationally under the brand name Zadaxin and is approved for use in more than thirty-five countries for the treatment of chronic hepatitis B and C and as an adjuvant to certain vaccines and chemotherapy regimens. While not FDA-approved in the United States, it holds four FDA Orphan Drug Designations. Mino supplies pharmaceutical-grade thymosin alpha 1 to licensed practitioners.

WHAT THE RESEARCH SHOWS

Thymosin alpha 1 has been studied in more than thirty published clinical trials. Research demonstrates its role in enhancing T-cell maturation and function, modulating dendritic cell activity, and restoring immune balance in patients with chronic viral infection, sepsis, and certain malignancies. It is one of the few compounds in this volume with a multi-decade clinical evidence base across major peer-reviewed journals.

CONSIDERED FOR

Use in immune-supportive protocols under physician supervision, particularly in the context of chronic viral conditions, post-illness recovery, and immune resilience strategies in longevity practices. Internationally established; FDA-approval in the U.S. remains pending.

BRAND	MECHANISM	STATUS
Zadaxin · SciClone	*T-cell maturation · TLR signaling*	*Approved · 35+ countries*

iv.

Cellular Longevity

The biology that sits one layer below everything else: the energetic machinery of the cell itself. Two compounds — one a coenzyme essential to every cell in the body, the other a peptide encoded by the mitochondrial genome it powers.

ENDOGENOUS COENZYME · COMPOUNDED

Mino NAD

NICOTINAMIDE ADENINE DINUCLEOTIDE

The most fundamental molecule in cellular energy metabolism — present in every cell of every living thing — and the cornerstone of the modern longevity conversation.

WHAT IT IS

NAD+, or nicotinamide adenine dinucleotide, is a coenzyme essential to every cell in the body. It plays a central role in oxidation-reduction reactions that power cellular energy production, and serves as a critical substrate for sirtuins and PARPs — enzymes involved in DNA repair, gene expression, and the regulation of cellular aging. NAD+ levels decline progressively with age, and restoring them is one of the most actively studied interventions in longevity science. Mino supplies pharmaceutical-grade NAD+ to licensed practitioners as an injectable formulation.

WHAT THE RESEARCH SHOWS

NAD+ biology has been the subject of thousands of peer-reviewed publications, with foundational work from laboratories at Harvard, MIT, and the Buck Institute. Research has linked NAD+ depletion to mitochondrial dysfunction, metabolic decline, and accelerated aging in preclinical models. Human trials of NAD+ precursors such as nicotinamide riboside and nicotinamide mononucleotide have shown the ability to raise circulating NAD+ levels safely. Direct intravenous and subcutaneous NAD+ protocols are widely used in clinical longevity practice, though large-scale human outcome trials remain ongoing.

CONSIDERED FOR

Use in physician-supervised cellular health and longevity protocols, often as part of broader metabolic and mitochondrial support strategies.

CLASS	MECHANISM	STATUS
Coenzyme · Pyridine nucleotide	*Sirtuin activation · Mitochondrial energy*	*Compounded · Investigational*

DISCOVERED 2015 · INVESTIGATIONAL

Mino MOTS

MOTS-C · MITOCHONDRIAL-DERIVED PEPTIDE

A peptide encoded not by the cell's nucleus, but by the mitochondria themselves — and a quiet rewriting of what we thought we knew about the geography of the genome.

WHAT IT IS

MOTS-c is a sixteen-amino-acid peptide encoded by a short open reading frame within the mitochondrial 12S rRNA gene. It was first characterized in 2015 by researchers at the University of Southern California and published in the journal Cell Metabolism. The discovery was significant because it confirmed that the mitochondrial genome — long thought to encode only a small set of proteins — also produces bioactive signaling peptides. Circulating MOTS-c levels decline with age. Mino supplies it as a research-grade peptide to licensed practitioners.

WHAT THE RESEARCH SHOWS

Preclinical research has shown that MOTS-c acts through the AMPK pathway to enhance skeletal muscle insulin sensitivity, regulate metabolic homeostasis, and influence nuclear gene expression related to stress adaptation and energy metabolism. In aged mice, MOTS-c administration has been shown to improve physical performance and metabolic function. Human clinical research is in early stages, with circulating levels of the peptide noted to be lower in patients with type 2 diabetes compared to healthy controls.

CONSIDERED FOR

Investigational use in research settings exploring mitochondrial health, exercise mimetic biology, and age-related metabolic decline.

DISCOVERED	MECHANISM	STATUS
2015 · *Lee et al.* · *Cell Metabolism*	*AMPK activation · Nuclear translocation*	*Investigational · Research only*

v.

The Endocrine Set

Four compounds covering the systems that shape how we look, how we think, and how we feel — growth and recovery, skin and collagen, the calm of the nervous system, and the brain biology of desire itself.

INVESTIGATIONAL · MOST-STUDIED GH PAIRING

Mino CJC · Ipamorelin

GHRH ANALOG · SELECTIVE GH SECRETAGOGUE

A two-peptide combination that mimics the body's natural rhythm of growth hormone release — and the most familiar protocol in the entire growth-hormone-supportive landscape.

WHAT IT IS

CJC-1295 is a long-acting analog of growth hormone-releasing hormone, modified to resist enzymatic degradation. Ipamorelin is a selective pentapeptide growth hormone secretagogue, originally characterized by Raun and colleagues in 1998 as the first GHS to release growth hormone without elevating cortisol or prolactin. Used in combination, the two compounds engage two distinct receptor pathways simultaneously, producing a synergistic effect on the body's pulsatile release of growth hormone. Neither compound is FDA-approved. Mino supplies them as research-grade peptides to licensed practitioners.

WHAT THE RESEARCH SHOWS

The foundational human study of CJC-1295, published by Teichman and colleagues in the Journal of Clinical Endocrinology and Metabolism in 2006, demonstrated dose-dependent increases in growth hormone of two- to ten-fold lasting six days or more, with corresponding elevations in IGF-1. Ipamorelin's selectivity profile makes it of particular interest for protocols aimed at restoring the natural physiology of growth hormone release without the off-target effects of older secretagogues.

CONSIDERED FOR

Physician-supervised use in protocols supporting recovery, sleep architecture, and body composition in adult patients, typically as part of broader hormone optimization strategies.

PIVOTAL STUDY	MECHANISM	STATUS
Teichman et al. · JCEM 2006	*GHRH-R + GHS-R1a synergy*	*Investigational · Off-label*

FIVE DECADES OF DERMATOLOGIC RESEARCH

Mino GHK

GHK-CU · GLYCYL-L-HISTIDYL-L-LYSINE COPPER

First isolated from human plasma in 1973 — and still, fifty years later, the most clinically validated copper-binding peptide in dermatology.

WHAT IT IS

GHK-Cu is a naturally occurring tripeptide composed of glycine, histidine, and lysine, complexed with a copper ion. It was originally isolated from human plasma by Pickart and Thaler in 1973, and circulating levels in the body decline significantly with age. The molecule's signature blue color comes from the copper coordination at its center. GHK-Cu has been studied for more than five decades and is one of the most extensively researched peptides in cosmetic and regenerative dermatology. Mino supplies pharmaceutical-grade GHK-Cu to licensed practitioners.

WHAT THE RESEARCH SHOWS

Multiple controlled clinical studies have shown that topical GHK-Cu increases collagen and elastin synthesis, improves skin density and thickness, and reduces fine lines and wrinkles. A landmark comparative trial reported collagen increases in 70% of women treated with GHK-Cu, outperforming both vitamin C and retinoic acid. Subsequent gene expression analyses have suggested GHK influences thousands of genes involved in tissue repair, inflammation control, and extracellular matrix remodeling.

CONSIDERED FOR

Use in dermatologic and aesthetic protocols supporting skin remodeling, post-procedure recovery, and the restoration of youthful collagen architecture.

DISCOVERED	MECHANISM	STATUS
1973 · *Pickart & Thaler*	*Collagen synthesis · MMP / TIMP balance*	*Cosmetic · Compounded*

RUSSIAN PHARMACOPOEIA · INVESTIGATIONAL IN U.S.

Mino Selank

SYNTHETIC TUFTSIN HEPTAPEPTIDE

A synthetic analog of a naturally occurring immune peptide — developed in Russia, used clinically there for more than two decades, and increasingly studied for its anxiolytic and cognitive properties.

WHAT IT IS

Selank is a synthetic heptapeptide developed at the Institute of Molecular Genetics of the Russian Academy of Sciences in collaboration with the V.V. Zakusov Institute of Pharmacology. It is based on the structure of tuftsin, a naturally occurring immunomodulatory peptide. Selank has been used clinically in Russia since the early 2000s as an anxiolytic, where it is part of the official pharmacopoeia. It is not approved by the FDA in the United States. Mino supplies it as a research-grade peptide to licensed practitioners.

WHAT THE RESEARCH SHOWS

Russian clinical research and a growing body of preclinical literature describe Selank's effects on the GABAergic and serotonergic systems, as well as its modulation of brain-derived neurotrophic factor expression. Studies report anxiolytic effects without the sedation or dependence associated with benzodiazepine therapy. Western clinical research remains limited, and most peer-reviewed evidence available in English comes from translated Russian publications and mechanistic studies.

CONSIDERED FOR

Investigational use in physician-supervised research settings exploring anxiolytic alternatives, cognitive support, and stress resilience.

ORIGIN	MECHANISM	STATUS
Russian Academy of Sciences	*GABAergic · BDNF modulation*	*Investigational · Research only*

FDA APPROVED · VYLEESI · 2019

Mino PT

BREMELANOTIDE · VYLEESI

The first and only FDA-approved peptide that works on the brain biology of sexual desire — and the closing compound of this volume.

WHAT IT IS

PT-141, known generically as bremelanotide and marketed under the brand name Vyleesi, is a synthetic cyclic heptapeptide derived from alpha-melanocyte-stimulating hormone. It is a melanocortin receptor agonist that activates MC3 and MC4 receptors in the central nervous system. The FDA approved Vyleesi in June 2019 as the first and only on-demand treatment for hypoactive sexual desire disorder in premenopausal women. Unlike vascular medications such as PDE5 inhibitors, PT-141 acts centrally on the neural pathways of desire itself. Mino supplies pharmaceutical-grade bremelanotide to licensed practitioners.

WHAT THE RESEARCH SHOWS

FDA approval was supported by two Phase 3 trials known as the RECONNECT studies, which together enrolled more than 1,200 premenopausal women with HSDD. The trials demonstrated statistically significant improvements in both sexual desire scores and reductions in the distress associated with low desire. Off-label investigational use in men, particularly those who do not respond fully to PDE5 inhibitors, is an active area of clinical interest.

CONSIDERED FOR

FDA-approved on-label use for premenopausal women with HSDD. Physician-supervised off-label use in other contexts of low desire and sexual dysfunction in both men and women.

APPROVED	MECHANISM	PIVOTAL TRIALS
FDA · June 2019	*MC3R / MC4R agonism · Central*	*RECONNECT 1 & 2*

A note on what this is — and isn't.

Every compound described in this volume is a regulated peptide product supplied by Mino Peptides exclusively to licensed medical practitioners. Nothing in these pages should be interpreted as medical advice, a treatment recommendation, or a suggestion that any of the compounds described are appropriate for self-administration.

Where compounds are described as FDA-approved, that approval is specific to the indications and patient populations defined in their official prescribing information. Where compounds are described as investigational, this means that they have not received FDA approval and are intended for research use under the supervision of a qualified physician. International approval status, where noted, refers to regulatory action in jurisdictions outside the United States and does not imply U.S. approval.

The scientific claims in this volume are drawn from peer-reviewed published literature current as of the 2026 edition. Patients interested in any of the therapies described should consult directly with a licensed physician.

Mino Peptides supplies practitioners. Practitioners care for patients. The distinction matters.

MINO PEPTIDES

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The most consequential science in medicine is rarely the loudest.

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